Task Analyzer
Phase 2 technical report

DRAFT - NOT READY FOR ACADEMIC SUBMISSION

Prepared on 2026-10-16 (America/Sao_Paulo). Repository owner: BudaBecker. The implementation and local checks are complete. Actual screenshots, human review, user merge and the required five-minute video remain pending.

Published implementation

Public repository: BudaBecker/sdd-task-analyzer

Phase 2 pull request #4 - open for review

Phase 2 source tree

Approved Phase 1 sources identified by the user (.specs/)

Delivered artifacts

Artifact	Purpose
src/task_analyzer.py	Typed Task records, analyze_tasks and TaskValidationError.
tests/test_harness.py	35 success, failure and boundary cases.
specs/task_analyzer_spec.md	Batch signature, reconciliation and acceptance examples.
CONTEXT_RULES.md	AI governance, evidence limits and approved boundaries.
pyproject.toml / uv.lock / requirements.txt	uv remains authoritative; requirements is an exported course artifact.

Contract reconciliation

The existing .specs/ features cover persistence and lifecycle. The Phase 2 contract explicitly adds the batch boundary; it does not claim those files already contained the challenge's numbered examples. Counts, elapsed duration and overdue eligibility preserve approved product examples. Priority grouping is an explicit academic input, not a new persistent task field. Numeric empty averages do not change the desktop's N/A presentation rule. The authoritative sources are .specs/ and docs/product/requirements.md. The brief's data_criacao and data_conclusao map to Task.created_at and Task.completed_at. REQ-023's deadline-window mean is excluded. Python 3.13+ satisfies the 3.11+ floor.

Submission guide, five-minute outline and remaining checklist

The correction session does not merge, deploy or modify a real database. Windows trailing-space zone acceptance remains a known limitation shared with the server; its zone-validation fix is a separate follow-up.

Verification and behavior

Actual local command results

Text record of executed checks, not a screenshot. The complete record is versioned in docs/phase-two/verification.txt. Tests use disposable paths; no real database was initialized or modified.

```
uv run python -m pytest tests/test_harness.py -v
35 passed in 0.07s

uv run python -m pytest -q
523 passed, 2 warnings in 81.59s (0:01:21)

uv run ruff format --check src tests
39 files already formatted

uv run ruff check src tests
All checks passed!

uv run python -m mypy --strict
Success: no issues found in 12 source files

uv pip check
All installed packages are compatible

uv build
Source distribution and wheel built successfully
```

The full suite consists of 487 existing server cases, 35 batch cases and one new installed-wheel batch smoke test outside the checkout. Two third-party deprecation warnings remain in Starlette/httpx and anyio integration.

Acceptance evidence

Scenario	Expected and verified outcome
1 - successful analysis	5 total, 3 completed, 2 pending; mean 3.0 hours; priority 4.0 and non-priority 2.5 hours; overdue rate 50.0%.
2 - invalid chronology	Completion before original creation raises TaskValidationError without a partial result or input mutation.
Empty / pending-only	Every average and rate is 0.0; counts retain their meaning.
Time and eligibility	UTC elapsed durations across DST; cutoff equality is late; pending/deleted tasks are excluded from the appropriate populations; Apia skipped-date rule retained.

These checks do not establish Ubuntu compatibility, Tailscale access, a metrics API or a completed Windows desktop application.

AI governance and completion

Prompt history and critical validation

The actual session instructions were to compare the intermediate challenge with the repository, execute the Phase 2 work with commits and pushes, and use .specs/ as the Phase 1 source. This is a summary, not a fabricated verbatim prompt transcript.

AI drafted the specification and CONTEXT_RULES.md during the initial session from the brief and .specs/ (commit 9bae22c, before code in 1c9daa1). The initial user prompts did not separately supply these newly created files as context. Their order alone does not demonstrate hallucinations were prevented.

An independent review found avoidable rate-precision loss, an invalid zone escaping as PermissionError, and tests that missed a .seconds duration defect. The correction session explicitly read the rules and spec before editing. It changed the rate to late * 100 / eligible, made its assertion exact, translated zone OSError and non-iterable inputs to TaskValidationError, and added multi-day and sub-second cases. Mutation checks now reject discarding days, rounding means and truncating seconds. Iterable-consumption errors still propagate.

The independent review is not the user's human PR review, which remains pending. Passing tests support executed cases, not every possible behavior.

Required final evidence

Item	Current status / next action
Terminal screenshot	Pending. Capture the actual harness command and 100%-passing summary.
Human review and merge	Pending. The user records their own review/comment in PR #4, then merges into main and retains the feature branch.
GitHub tree screenshot	Pending. After merge, verify and capture the required artifacts on main; the default branch is currently incomplete.
Pull request screenshot	Pending. Capture PR #4 with its changes and verification. Human review remains pending.
Five-minute YouTube video	Pending. Record horizontally, with visible camera and audio; use Unlisted or Public visibility, not a Short. No URL has been supplied.
Required Phase 1 Docs link	Pending (rubric, page 6). Include the actual Google Docs URL as a link only; do not reimport its content. Add required student details.
Final PDF / salaonline	After user merge, generate the final PDF with repository root, tree/main, PR #4 and required Docs/video links; review and submit. Not performed.